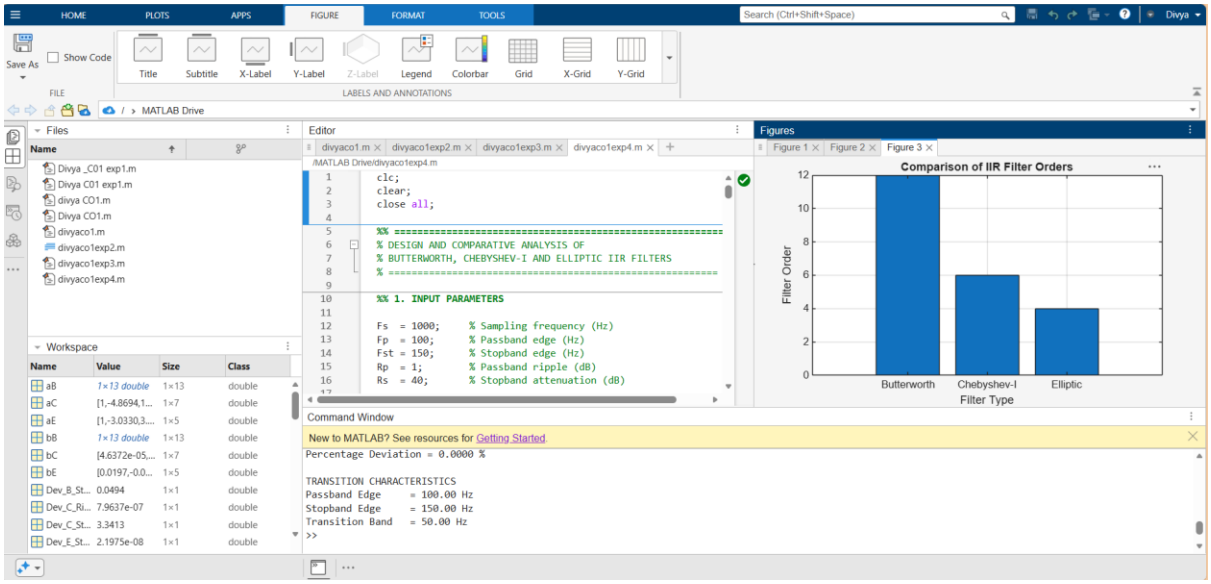
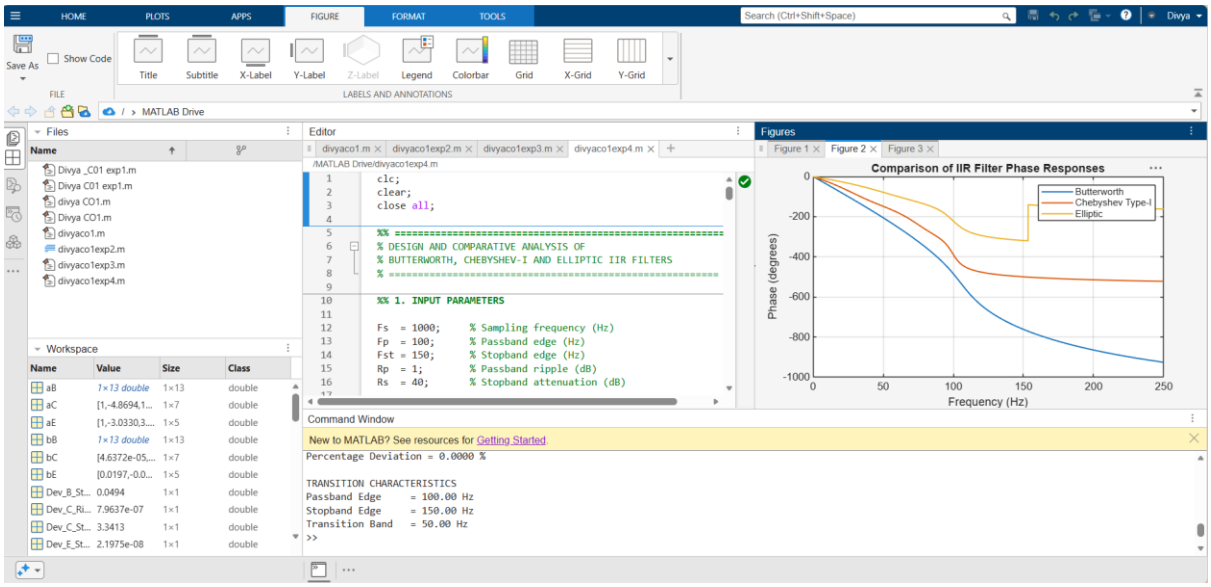
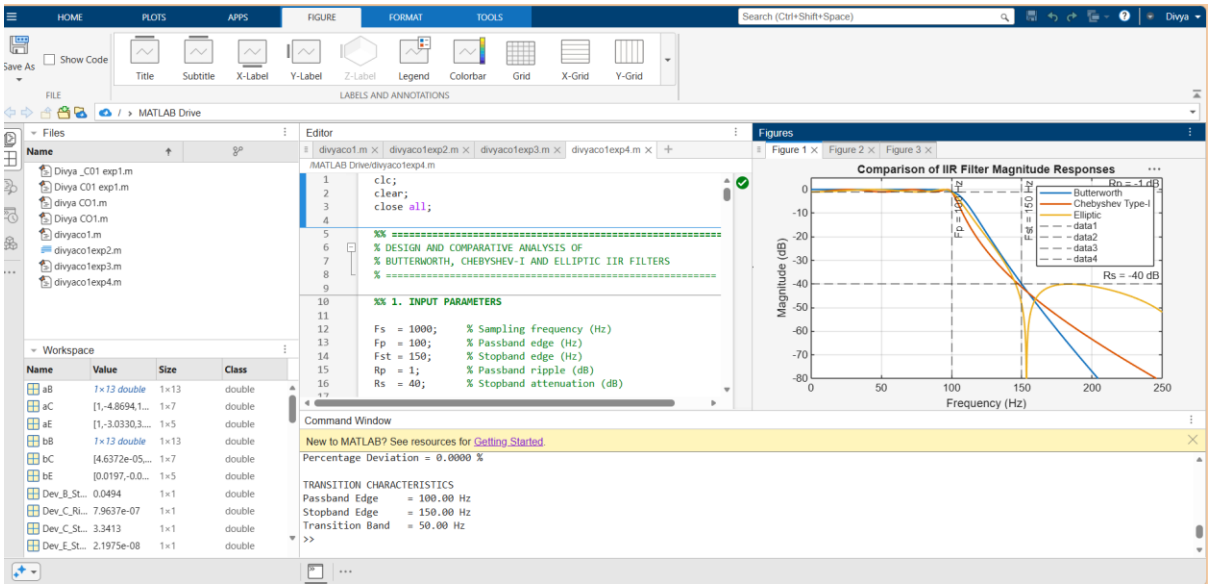

HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Divya
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
FILE
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
Divya _C01 exp1.m
Divya C01 exp1.m
divya CO1.m
Divya CO1.m
divyaco1.m
divyaco1exp2.m
divyaco1exp3.m
divyaco1exp4.m
Workspace
Name
Value
Size
Class
Editor
divyaco1.m
divyaco1exp2.m
divyaco1exp3.m
divyaco1exp4.m
/MATLAB Drive/divyaco1exp4.m
clc;
clear;
close all;
%% ==
% DESIGN AND COMPARATIVE ANALYSIS OF
% BUTTERWORTH, CHEBYSHEV-I AND ELLIPTIC IIR FILTERS
% ==
%% 1. INPUT PARAMETERS
Fs = 1000; % Sampling frequency (Hz)
Fp = 100; % Passband edge (Hz)
Fst = 150; % Stopband edge (Hz)
Rp = 1; % Passband ripple (dB)
Rs = 40; % Stopband attenuation (dB)
Figures
Figure 1
Figure 2
Figure 3
Comparison of IIR Filter Orders
Filter Order
Butterworth
Chebyshev-I
Elliptic
Filter Type
Command Window
New to MATLAB? See resources for Getting Started.
Percentage Deviation = 0.0000 %
TRANSITION CHARACTERISTICS
Passband Edge = 100.00 Hz
Stopband Edge = 150.00 Hz
Transition Band = 50.00 Hz
>>


HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Divya
Save As
Show Code
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
FILE
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
Divya _C01 exp1.m
Divya C01 exp1.m
divya CO1.m
Divya CO1.m
divyaco1.m
divyaco1exp2.m
divyaco1exp3.m
divyaco1exp4.m
Workspace
Name
Value
Size
Class
Editor
divyaco1.m
divyaco1exp2.m
divyaco1exp3.m
divyaco1exp4.m
/MATLAB Drive/divyaco1exp4.m
clc;
clear;
close all;
%% ==
% DESIGN AND COMPARATIVE ANALYSIS OF
% BUTTERWORTH, CHEBYSHEV-I AND ELLIPTIC IIR FILTERS
% ==
%% 1. INPUT PARAMETERS
Fs = 1000; % Sampling frequency (Hz)
Fp = 100; % Passband edge (Hz)
Fst = 150; % Stopband edge (Hz)
Rp = 1; % Passband ripple (dB)
Rs = 40; % Stopband attenuation (dB)
Figures
Figure 1
Figure 2
Figure 3
Comparison of IIR Filter Phase Responses
Butterworth
Chebyshev Type-I
Elliptic
Phase (degrees)
Frequency (Hz)
Command Window
New to MATLAB? See resources for Getting Started.
Percentage Deviation = 0.0000 %
TRANSITION CHARACTERISTICS
Passband Edge = 100.00 Hz
Stopband Edge = 150.00 Hz
Transition Band = 50.00 Hz
>>


HOME
PLOTS
APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Divya
Save As
Show Code
FILE
Title
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
MATLAB Drive
Files
Name
Divya _C01 exp1.m
Divya C01 exp1.m
divya CO1.m
Divya CO1.m
divyaco1.m
divyaco1exp2.m
divyaco1exp3.m
divyaco1exp4.m
Workspace
Name
Value
Size
Class
aB 1×13 double 1×13 double
aC [1,-4.8694,1... 1×7 double
aE [1,-3.0330,3.... 1×5 double
bB 1×13 double 1×13 double
bC [4.6372e-05,... 1×7 double
bE [0.0197,-0.0... 1×5 double
Dev_B_St... 0.0494 1×1 double
Dev_C_Ri... 7.9637e-07 1×1 double
Dev_C_St... 3.3413 1×1 double
Dev_E_St... 2.1975e-08 1×1 double
Editor
divyaco1.m
divyaco1exp2.m
divyaco1exp3.m
divyaco1exp4.m
/MATLAB Drive/divyaco1exp4.m
clc;
clear;
close all;
%% ==
% DESIGN AND COMPARATIVE ANALYSIS OF
% BUTTERWORTH, CHEBYSHEV-I AND ELLIPTIC IIR FILTERS
% ==
%% 1. INPUT PARAMETERS
Fs = 1000; % Sampling frequency (Hz)
Fp = 100; % Passband edge (Hz)
Fst = 150; % Stopband edge (Hz)
Rp = 1; % Passband ripple (dB)
Rs = 40; % Stopband attenuation (dB)
Figures
Figure 1
Figure 2
Figure 3
Comparison of IIR Filter Magnitude Responses
Rp = -1 dB
Fp = 100 Hz
Fst = 150 Hz
Butterworth
Chebyshev Type-I
Elliptic
data1
data2
data3
data4
Rs = -40 dB
Magnitude (dB)
Frequency (Hz)
0
-10
-20
-30
-40
-50
-60
-70
-80
0
50
100
150
200
250
Command Window
New to MATLAB? See resources for Getting Started.
Percentage Deviation = 0.0000 %
TRANSITION CHARACTERISTICS
Passband Edge = 100.00 Hz
Stopband Edge = 150.00 Hz
Transition Band = 50.00 Hz
>>